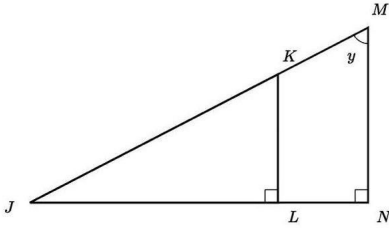


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1.



Note: figure not drawn to scale.

In the figure, JLK and JNM are both right triangles. Which of the following is equal to KL ?

Choose 1 answer:

(A) $\frac{JK \cdot \sin(y)}{JL}$

(B) $\frac{JK}{JL \cdot \cos(y)}$

(C) $JL \cdot \tan y$

(D) $\frac{JL}{\tan y}$

2.

The equation of a parabola is written in the form $y = ax^2 + bx + c$, where a , b , and c are constants.

When graphed in the xy -plane, the parabola has vertex $(-1, 4)$ and does not intersect the x -axis. Which of the following could be the value of $a - b - c$?

(A) -5

(B) -4

(C) 0

(D) 4

3.

In the xy -plane, a parabola has a vertex $(7, 12)$ and intersects the x -axis at two points. If the equation of the parabola is written in the form $y = ax^2 + bx + c$, where a , b , and c are constants, which of the following could be the value of $a + b + c$?

A) 10

B) 14

C) 18

D) 36

4.

An environmental scientist is investigating the volatility of 175 organic compounds by finding the boiling point of each compound. The mean boiling point of all 175 compounds is 184 degrees Celsius. The scientist classifies each of these compounds as either semivolatile or volatile. Of the 175 compounds, 50 compounds were classified as semivolatile, and these 50 compounds have a mean boiling point of 339 degrees Celsius. The remaining 125 compounds were classified as volatile. What is the mean boiling point, in degrees Celsius, of the 125 compounds classified as volatile compounds?

5.

$$5x + 7y = 1$$

$$ax + by = 1$$

In the given pair of equations, a and b are constants. The graph of this pair of equations in the xy -plane is a pair of perpendicular lines. Which of the following pairs of equations also represents a pair of perpendicular lines?

A) $10x + 7y = 1$

$$ax - 2by = 1$$

B) $10x + 7y = 1$

$$ax + 2by = 1$$

C) $10x + 7y = 1$

$$2ax + by = 1$$

D) $5x - 7y = 1$

$$ax + by = 1$$

6.

Sample	Percent in favor	Margin of error
<i>A</i>	67.83%	7.8%
<i>B</i>	72.94%	4.5%

The results of two random samples of votes for a proposition are given in the table. The samples were selected from the same population, and the margins of error were calculated using the same method.

Which of the following is the most appropriate reason that the margin of error for sample *A* is greater than the margin of error for sample *B*?

- A) Sample *A* had a larger sample size.
- B) Sample *A* had a smaller sample size.
- C) Sample *A* had a smaller number of votes that could not be recorded.
- D) Sample *A* had a lower percent of favorable responses.

7.

There are red, blue, green, and black marbles in a bag. There are 26 black marbles, 8 green marbles, and 26 blue and red marbles. Given that a randomly selected marble is not black, the probability of selecting a red marble is $\frac{5}{17}$. What is the number of blue marbles in the bag?

8.

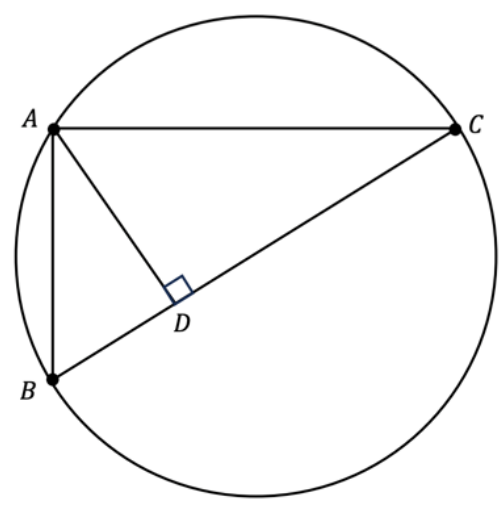
For the given parabola $ax^2 + bx + c$, a , b , and c are constants. The x -intercepts of the parabola are at $(14, 0)$ and $(k, 0)$. If $f(27) = f(11)$, what is the value of k ?

9.

There are two ranches. Ranch *A* and ranch *B*. The ratio of male horses to female horses is 1: 15 for the total number of horses on ranch *A* and *B*. The ratio of female horses on ranch *A* to ranch *B* is 7: 17. There are 168 horses on ranch *A* and 344 horses on Ranch *B*. How many more male horses are there on ranch *A* than on ranch *B*?

10.

In the given figure, BC is the diameter of the circle. If the length of BC is equal to 132 and the length of AB is equal to $\sqrt{363}$, what is the value of $\frac{BC}{BD}$?



Note: Figure not drawn to scale.

11.

The table gives the volume and surface areas of two similar rectangles, where k is a constant.

	Volume (cubic inches)	Surface Area (square inches)
Rectangle A	4,374	1,782
Rectangle B	1,500,282	k

What is the value of k ?

- A) 5,346
- B) 12,474
- C) 87,318
- D) 611,226

12.

$$f(x) = (x + a)(x + b)$$

For the given function $f(x)$, where a and b are integer constants, $f(-21) > 0$, $f(-15) > 0$, and $f(-18) < 0$, what is one possible value of $a + b$?

13.

The mass of particles A , B , and C are a atomic mass units, b atomic mass units, and c atomic mass units, respectively. If the mass of particle A is 4,600% of the mass of particle C , and the mass of particle C is 0.004% of the mass of particle B , which expression represents the value of $a + b$ in terms of c ?

- A) $2,960c$
- B) $4,604c$
- C) $4,850c$
- D) $25,046c$

14.

A circle with a center $B(0, 0)$ is a unit circle on the xy -plane. Point $A(1, 0)$ and C lie on the circle. If the measurement of angle ABC is $\frac{975\pi}{60}$ in radians, what is the x -coordinate of point C ?

- A) $\frac{\sqrt{2}}{2}$
- B) $-\frac{\sqrt{2}}{2}$
- C) 0
- D) $\frac{\sqrt{2}}{3}$

15.

Which of the following has $x + 2a$ as a factor where a is a positive integer?

- A) $3x^2 + 7x + 14a = 0$
- B) $3x^2 + 28x + 14a = 0$
- C) $3x^2 + 42x + 14a = 0$
- D) $3x^2 + 49x + 14a = 0$

16.

The function f is defined by $f(x) = ax^2 + bx + c$, where a , b and c are constants. The graph of $y = f(x)$ in the xy -plane passes through the points $(13, 0)$ and $(-6, 0)$. Which of the following is the value of $a + b$ in terms of a ?

- A) $7a$
- B) $6a$
- C) $-6a$
- D) $-7a$

17.

In triangle ABC , angle C is a right angle, point D lies on $\overline{AB}$, point E lies on $\overline{BC}$, and $\overline{DE}$ is parallel to $\overline{AC}$. If the length of $\overline{AB}$ is 39 units, $\overline{DE}$ is 8 units, the length of $\overline{AC}$ is greater than the length of $\overline{BC}$, and the area of triangle ABC is 270 units, what is the length of $\overline{BE}$, in units?

18.

A scientist is creating a sugar water solution for his lab. If there are currently 10 liters of a 32% sugar solution in a bucket, how many liters of a 12% sugar solution need to be added to the bucket to create a 20% sugar solution??

19.

Bob's Woodshop buys 2 tons of wood from The Wood Depot. Each ton of wood costs \$2,500. Bob gets a 15% discount for being a frequent customer. After the discount is applied, Bob has to pay a 10% sales tax. How much did Bob pay for the 2 tons of wood?

20.

The product of three numbers is 2160. One of the numbers, z , is 25% greater than the other two numbers, which are the same. What is the value of z ?

21.

If the vertex of a parabola is at the point $(1, 24)$ and the quadratic can be written in the form $-a(x^2) + bx + c$ and a, b , and c are all positive integers, what is the maximum value of b ?

22.

A boxing club membership costs \$85 per month. After 1 year, members receive a discount of \$30 off the cost of their monthly membership. Which function c gives the total cost $c(t)$, in dollars, that a new member pays after t months of membership, where $t \geq 12$?

- A) $c(t) = 1,020 + 55t$
- B) $c(t) = 55t + 85$
- C) $c(t) = 360 + 55t$
- D) $c(t) = 1,020 + 85(t - 12)$

23.

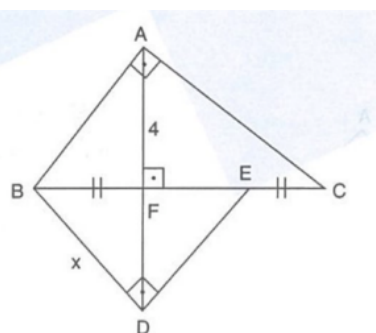
A box contains 1 red bead, 6 white beads, and 3 blue beads. Jorge will randomly remove one bead from the box, record its color, and place it back in the box. If Jorge repeats this experiment 150 times, what is the expected number of times that Jorge will record a bead that is blue?

24.

A study conducted by a physician found that the average number of hours 11th graders in high school spent sleeping during weeknights was 7.2, with an associated margin of error of 0.7 hours. The study was then repeated with a much smaller sample size, with the mean and margin of error of the new sample being calculated in the same way as the original study. Which of the following is most likely true?

- A) The margin of error of the new study is larger than the margin of error of the original study.
- B) The margin of error of the new study is smaller than the margin of error of the original study.
- C) The mean from the new study is larger than the mean from the original study.
- D) The mean from the new study is smaller than the mean from the original study.

25.



In the picture above $AF=4$ and $BF=EC$
What is the value of x

26. A quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object was launched. According to the model, the object was launched from a height of 0 feet and reached its maximum height of 1,600 feet 10 seconds after it was launched. Based on the model, what was the height, in feet, of the object 15 seconds after it was launched?

27.

- For the positive quantities h , j , and k , 94% of h is equivalent to 47% of j , and j is equivalent to 30% of k . What percentage of k is h ? (Disregard the % sign when entering your answer. For example, if your answer is 39%, enter 39)

The manager of a gym selected a sample of 139 members at random to estimate the percentage of the gym's members that would continue to pay for a membership if the price increased. From the survey, the manager estimates that 82% of the gym's members would continue to pay for a membership if the price increased, with an associated margin of error of 6.39%. If the survey is repeated with a random sample of 278 members and the results are calculated in the same way, which of the following will be the most likely effect of using the larger random sample compared to the smaller random sample?

- (A) The margin of error will be lower.
- (B) The margin of error will be higher.
- (C) The estimate of the percentage of the gym's members that would continue to pay for a membership if the price increased will be lower.
- (D) The estimate of the percentage of the gym's members that would continue to pay for a membership if the price increased will be higher.

29.

For groups of 25 or more people, a museum charges \$28 per person for the first 25 people and \$17 for each additional person. Which function f gives the total charge, in dollars, for a tour group with n people, where $n \geq 25$?

(A) $f(n) = 17n + 275$

(B) $f(n) = 17n + 28$

(C) $f(n) = 17n + 700$

(D) $f(n) = 45n - 425$

30.

The percent increase in mass of a certain red kangaroo from 80 days old to 160 days old was 676%. If this red kangaroo's mass was k grams at 80 days old, which expression represents its mass, in grams, at 160 days old?

(A) $7.76k$

(B) $6.76k$

(C) $1.08k$

(D) $0.08k$

31.

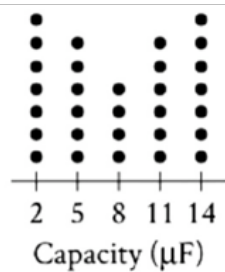
In triangle FGH and triangle KLM , the measures of angles G and L are each 30° . The lengths of $\overline{GH}$ and $\overline{LM}$ are each 34 centimeters, and $\frac{GH}{GF} = \frac{LM}{LK}$. Which additional piece of information would be necessary to prove that triangle FGH is congruent to triangle KLM ?

(A) The lengths of $\overline{FG}$ and $\overline{KM}$ are each 17 centimeters.

(B) The measures of angles H and M are each 60° .

(C) The measures of angles F and K are each 90° .

(D) No additional information is necessary.



The dot plot shows the distribution of capacity for a set of capacitors, set A, which a researcher used for a certain experiment. For another experiment, the researcher used a different set of capacitors, set B. Set B has the same number of capacitors as set A but the capacity of each capacitor of set B is **17 microfarads (μF)** greater than the capacity of each respective capacitor in set A. Which of the following is true about the capacities of the capacitors in set B?

- (A) The mean capacity is $8 \mu\text{F}$, and the range of capacities is $12 \mu\text{F}$.
- (B) The mean capacity is $8 \mu\text{F}$, and the range of capacities is $29 \mu\text{F}$.
- (C) The mean capacity is $25 \mu\text{F}$, and the range of capacities is $12 \mu\text{F}$.
- (D) The mean capacity is $25 \mu\text{F}$, and the range of capacities is $29 \mu\text{F}$.

33.

The mass of object A is 498% of the mass of object B, and the mass of object A is 0.830% of the mass of object C. If the mass of object C is $p\%$ of the mass of object B, what is the value of p ?

34.

A right triangular pyramid has exactly six edges. Each of these edges is 96 centimeters long. If the surface area of the pyramid is $k\sqrt{3}$ square centimeters, what is the value of k ?

35.

The functions f and g are defined by the given equations below, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?

I. $f(x) = 18(1.4)^{x+5}$

II. $g(x) = 18(1.16)(1.4)^{x+3}$

(A) I only

(B) II only

(C) I and II

(D) Neither I nor II

36.

The positive number a is 2,352% of the sum of the positive numbers b and c , and b is 84% of c . What percent of b is a ?

(A) 5,152%

(B) 2,436%

(C) 51.52%

(D) 24.36%

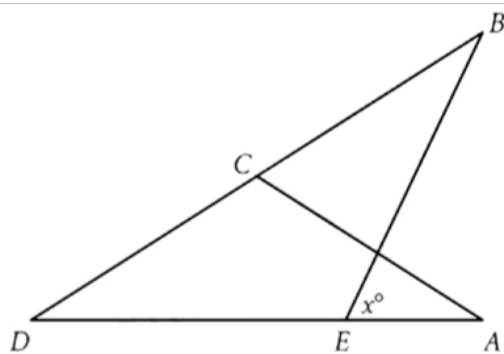
37.

Line j is defined by $4x + 5y = 55$. Line k is parallel to line j in the xy -plane. An equation of line k is $24x + ry = 15$, where r is a constant. If line k passes through the point $(0, b)$, what is the value of b ?

38.

Function f is a quadratic function where $f(-20) = 0$ and $f(-4) = 0$. The graph of $y = f(x)$ in the xy -plane has a vertex at $(r, -64)$. What is the value of r ?

39.



Note: Figure not drawn to scale.

In the figure, $AC = CD$. The measure of angle EBC is 29° , and the measure of angle ACD is 108° . What is the value of x ?

40.

The town of Fillmore is planning its annual fireworks show. The equation $y = -16(x - 6.8)^2 + 740$ gives the estimated height y , in feet, of a type of firework x seconds after it is launched into the air. Which of the following is the best interpretation of the vertex of the graph of the equation in the xy -plane?

- (A) This type of firework reaches an estimated maximum height of 740 feet 16 seconds after it is launched into the air.
- (B) This type of firework reaches an estimated maximum height of 740 feet 6.8 seconds after it is launched into the air.
- (C) This type of firework reaches an estimated maximum height of 16 feet 740 seconds after it is launched into the air.
- (D) This type of firework reaches an estimated maximum height of 6.8 feet 740 seconds after it is launched into the air.

41.

The function f is defined by $f(x) = ax^2 + bx + c$, where a , b , and c are constants. The graph of $y = f(x)$ in the xy -plane passes through the points $(8, 0)$ and $(-2, 0)$. If a is an integer greater than 1, which of the following could be the value of $a + b$?

- (A) -10
- (B) -5
- (C) 6
- (D) 7

42.

$7x + 5y = 2$

The given equation is one equation in a system of two linear equations. If the system of equations has at least one solution, which of the following equations could be the other equation in the system?

- I. $10.5x + 7.5y = 3$
- II. $10.5x - 7.5y = 3$

- (A) I only
- (B) II only
- (C) I and II
- (D) Neither I nor II

43.

For the function f , for every increase of 2 in the value of x , the value of $f(x)$ increases by a factor of c , where c is a constant. Which of the following equivalent forms of function f displays the value of c as the base or the coefficient?

- (A) $f(x) = 48(2)^{6x}$
- (B) $f(x) = 48(8)^{2x}$
- (C) $f(x) = 48(64)^x$
- (D) $f(x) = 48(4,096)^{\frac{x}{2}}$

44.

A bin contains a mixture of T-shirts for two sports teams. The table shows the number of T-shirts in the bin, classified by size and sports team.

	Flames	Sharks	Total
Small	8	7	15
Medium	22	34	56
Large	14	27	41
Total	44	68	112

One T-shirt from the bin will be selected at random. What is the probability of selecting a T-shirt that is medium, given that the T-shirt is a Flames T-shirt?